# JONATHAN TOBIAS DA SILVA

## Senior Software Engineer, M.Sc. | Embedded Software and Systems

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Senior Software Engineer specializing in embedded software and systems, focusing on operational and information technology domains. Dedicated to leveraging technology and engineering expertise to address real-world challenges and drive meaningful progress.

## EXPERIENCE

### Senior Embedded Software Engineer

**Act Digital**

2026 – Ongoing | Remote

- Develop and customize mission-critical embedded software for Embedded Linux and RTOS (VxWorks), focusing on industrial switches and PLCs.
- Adapt software solutions to meet strict industrial and substation requirements, ensuring high availability, determinism, and peak performance.
- Implement and maintain industrial communication protocols, including OPC UA, EtherNet/IP, and Modbus, as well as TCP/IP and UDP network stacks.
- Work closely with hardware teams on device bring-up, driver integration, and software-hardware productization for global markets.
- Optimize real-time system behavior and latency to meet the rigorous standards of Industrial IoT (IIoT) and critical infrastructure.
- Collaborate with cross-functional global teams (U.S. and Brazil), applying DevOps practices, CI/CD, and advanced troubleshooting to ensure software excellence.

### R&D Embedded Software Engineer

**Datacom**

2024 – 2026 | Remote

- Develop and maintain C/C++ and Python software, optimizing device performance and ensuring high reliability.
- Enhance code reliability through the design and execution of unit tests and automated integration tests using GoogleTest and the Robot Framework.
- Develop and debug embedded Linux systems, with a focus on performance tuning and system stability across various hardware platforms.
- Implement and maintain protocols at the Layer 3 level.
- Conduct code reviews following best practices in TDD, CI/CD, and Clean Code principles.
- Conduct in-depth troubleshooting on complex telecommunications systems, resolving critical issues and minimizing system downtime.

### R&D Electronic Intelligence Analyst

**Nova Smar**

2023 – 2024 | Sertãozinho, SP, Brazil

- Coordinate a product development initiative, managing a team of 10 engineers and developers.
- Apply cybersecurity measures to ensure compliance with the IEC 62443 and O-PAS standards, enhancing the security and integrity of embedded systems.
- Orchestrate the migration of version control systems from CVS to Git for the R&D department, implementing CI/CD pipelines to streamline project workflows.
- Lead the development and maintenance of Linux operating systems and kernels using the Yocto Project for real-time operating systems (RTOS).

## EDUCATION

### Master in Electrical Engineering

**University of São Paulo (USP)**

2022 – 2025 | São Carlos, SP, Brazil

**Thesis:**

- Vulnerability analysis of industrial OPC UA networks.

### Bachelor in Electrical Engineering

**Federal Institute of São Paulo (IFSP)**

2017 – 2021 | Sertãozinho, SP, Brazil

**Final Paper:**

- Cloud-based intelligent system for the detection of hydraulic pump failures.

### Certificate in Administration

**ETEC Antônio de Pádua Cardoso**

2014 – 2016 | Batatais, SP, Brazil

- Acquired foundational technical knowledge essential for business and people management.

## TOPICS OF INTEREST

Embedded Systems | Convergence IT/OT | Linux | Cybersec | Cloud Computing | AI/ML | Management | Software Engineering | Communication Protocols | Programming Languages | Data Science & Analystics

## PUBLICATIONS

### Journal Articles

- **J. T. Da Silva** and O. Paschoal, "Fortifying industrial automation: Best practices in cybersecurity," 2023.

### Conference Proceedings

- **J. T. Da Silva**, A. L. Dias, and I. N. Da Silva, "A survey on opc ua protocol: Overview, challenges and opportunities," in *15th IEEE International Conference on Industry Applications (INDUSCON)*, 2023.

- Develop desktop and embedded software applications and internal tools to optimize workflows.
- Experience with hardware interfaces, networking protocols, and communication standards.

### R&D Electronic Intelligence Assistant

**Nova Smar**

2022 – 2023 Sertãozinho, SP, Brazil

- Research Cybersecurity and Artificial Intelligence for Industrial Automation, ensuring system security and efficiency gains of 76% in production processes through Machine Learning tools.
- Develop desktop and embedded applications for System302 in Linux environments, including firmware patches that have enabled updates for the largest installed base of the system.
- Apply a comprehensive set of Software Engineering and Computer Network concepts to development processes such as version control, clean code, and CI/CD.
- Manage System302 project milestones and quality assurance processes to ensure project success.
- Craft technical documentation in Portuguese and English.
- Develop and conduct internal training.

R&D

### Intern

**Nova Smar**

2021 Sertãozinho, SP, Brazil

- Experience in software development, research, testing, documentation, technical support, and software management within R&D.

- A. L. Dias, **J. T. Da Silva**, A. C. Turcato, and G. S. Sestito, "An intelligent fault diagnosis for centrifugal pumps based on electric current information available in industrial communication networks," in *14th IEEE International Conference on Industry Applications (INDUSCON)*, 2021.
- **J. T. Da Silva**, A. L. Dias, N. F. D. F. Filho, and A. C. Turcato, "Development of a web-based tool for dimensioning equipment for electric motor drives," in *10th IFSP Scientific and Technological Initiation Congress (CONICT)*, 2020.

## SKILLS

### Technical

- **Computer technologies and programming languages:** C, C++, Python, ShellScript, (Embedded) Linux, Git, Docker, LaTeX, Qt (+QML), Database (MongoDB, MySQL, PostgreSQL);
- **R&D methodologies and techniques:** TDD, DevOps (CI/CD, Automation), FMEA;
- **Experience with agile methods and tools:** OKR, SCRUM, Kanban, Gerrit, GitHub, GitLab, Bugzilla, Jenkins, Mantis Bug Tracker, Atlassian Tools, DotProject.

### Professional

- High leadership ability;
- High adaptability;
- Deadline-Oriented;
- Eagerness for knowledge;
- Excellent teamwork skills;
- Always ready for a collaborative environment.